Interactions of Emerin with Histone-Modifying Enzymes in Nuclear Envelope-Mediated Repression

Aayush Gupta[1,2], Richard T. Condon[2,3], and James M. Holaska[2,3]

[1] Edison High School, [2] Department of Biomedical Sciences, Cooper Medical School of Rowan University, Camden, New Jersey, US

[3] Rowan-Virtua School of Translational Biomedical Engineering and Sciences, Stratford, New Jersey, USA

Introduction

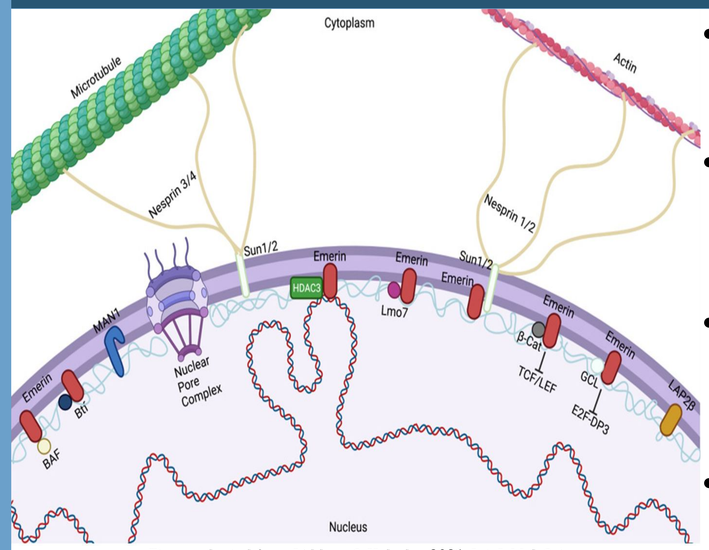


Figure adapted from Liddane & Holaska, 2021, Int. J. Mol. Sci.

Figure 1

- **Nuclear envelope:** a double-membrane structure enclosing the genome, lined internally by a meshwork of lamin proteins called the lamina.
- **Emerin:** an inner nuclear membrane protein that anchors chromatin to the nuclear periphery and recruits histone-modifying enzymes to regulate gene expression.
- **Histone-modifying enzymes (HDAC3, EZH2, G9a):** add or remove chemical marks on histones to condense or loosen chromatin, controlling whether nearby genes are silenced or active.
- **Chromatin organization:** genes near the nuclear periphery tend to be silenced; genes in the nuclear interior tend to be

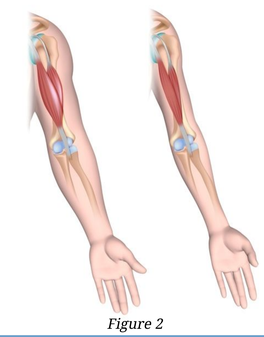


Figure 2

- **Emery-Dreifuss Muscular Dystrophy (EDMD):** a rare disorder causing progressive muscle wasting and cardiac conduction defects, caused by reduced emerin expression.
- Loss of emerin function disrupts chromatin organization and impairs muscle cell differentiation.

We **hypothesize** that dynamic interactions between **emerin** and **histone modifying enzymes** control the repositioning of crucial **myogenic genes** at the inner nuclear membrane during cellular differentiation.

Methods

- **Immunofluorescence & Confocal/STED Microscopy:** visualize the localization of emerin and histone-modifying enzymes within the nucleus.
- **Co-Immunoprecipitation (Co-IP):** isolates emerin from a cell lysate along with any proteins physically bound to it.
- **SDS-PAGE:** separates proteins by molecular weight via gel electrophoresis.
- **Western Blot:** transfers separated proteins to a membrane and probes for specific proteins using antibodies.
- **Cell lines used:** HeLa cells and C2C12 myogenic progenitor cells, differentiated over a 48-hour time course.

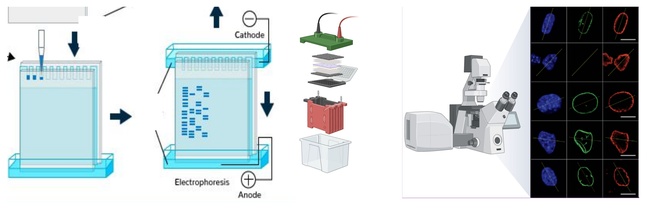


Figure 3

Results (STED Imaging)

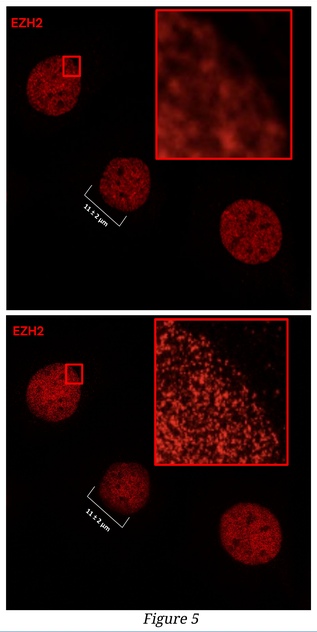


Figure 5

Confocal Microscopy

Nikon Widefield Live Cell Imaging Rig

11 ± 2 µm

STED Microscopy

Abberior STEDYCON Superresolution STED and Confocal Microscope

11 ± 2 µm

Results

(Immunofluorescence & Co-Immunoprecipitation)

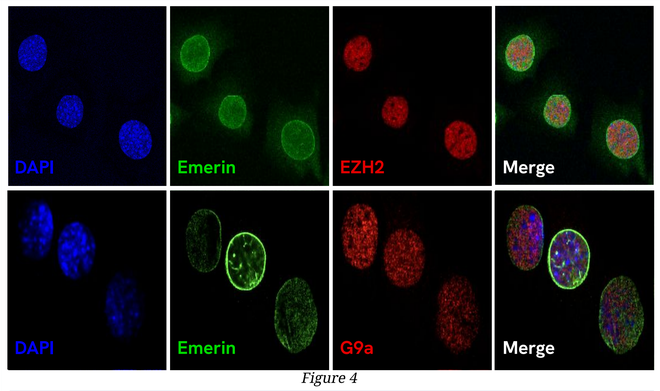


Figure 4

Immunofluorescence shows Emerin (green) localized to the nuclear rim, while EZH2 and G9a (red) are distributed diffusely throughout the nucleus. Merged images show only minimal overlap between emerin and either enzyme at the nuclear periphery.

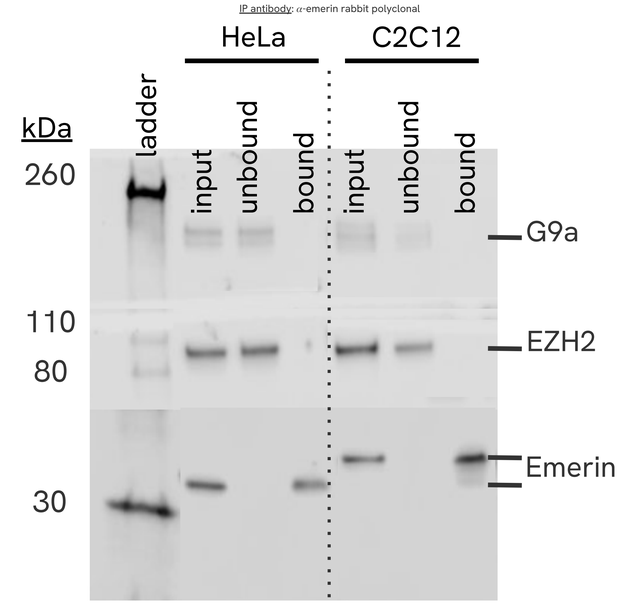


Figure 6

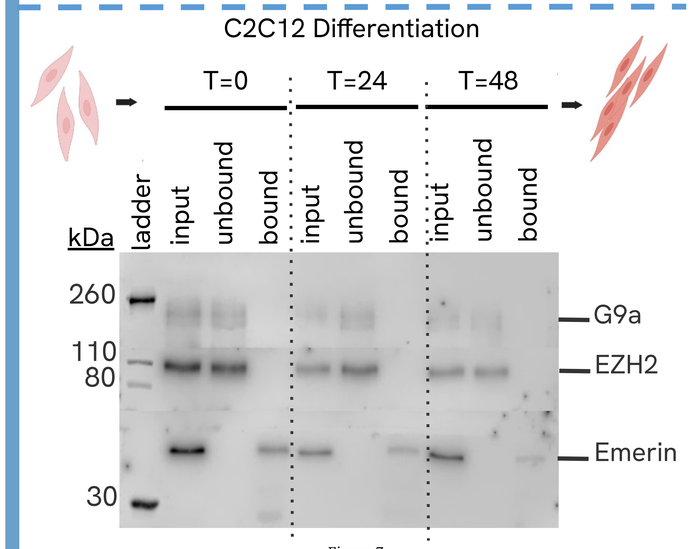
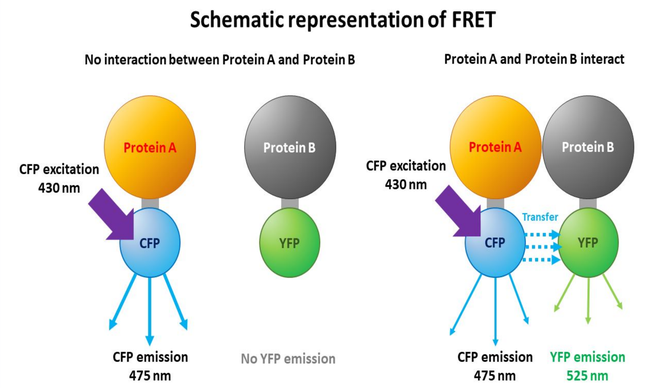


Figure 7

Conclusions/Future Directions



Berthold Technologies GmbH & Co. KG. (n.d.). *Fluorescence resonance energy transfer (FRET)* [Illustration]. Berthold Technologies.

Figure 8

- Standard immunofluorescence did not reveal co-localization between emerin and EZH2 or G9a at the nuclear periphery.
- Super-resolution STED imaging confirmed this result, ruling out limited resolution as the cause.
- Co-immunoprecipitation recovered emerin but not EZH2 or G9a, suggesting these interactions are too transient to capture biochemically.
- Future Direction: We plan to use Förster Resonance Energy Transfer (FRET) to detect these fast, transient interactions in real time, since FRET only signals when two proteins are within 1–10 nm of each other.

References

- Liddane, S. & Holaska, J.M. (2021). *Int. J. Mol. Sci.*
- Marano, N. & Holaska, J.M. (2025). *The FASEB Journal.*
- Marano, N. & Holaska, J.M. (2022). Emerin interacts with histone methyltransferases to regulate repressive chromatin at the nuclear periphery. *Front Cell Dev Biol. 10:1007120.*
- Schmidt, M. et al. (2019). *Cell. Mol. Life Sci.*

Acknowledgements

Sincere Thanks to:

Dr. Holaska
Richard Condon
Christal Rolling
Emily Hansen
Dr. Vega
Myranda Sims

ROWAN-VIRTUA School of Translational Biomedical Engineering & Sciences

National Institutes of Health

Cooper Medical School of Rowan University